

①

DATE 17 01 2024

Matter

- 18 As the readings are at same time, the angle of elevation of sun is same.

$$\angle ABC = \angle DFE = \angle PRQ \rightarrow \textcircled{1}$$

i In $\triangle ABC$ & $\triangle DFE$

$$\angle ABC = \angle DFE \text{ (from } \textcircled{1})$$

$$\angle ACB = \angle DEF (90^\circ)$$

By AA similarity

$$\triangle ACB \sim \triangle DFE$$

$$\frac{AC}{DE} = \frac{CB}{EF} = \frac{AB}{DF}$$

$$\frac{20}{DE} = \frac{10}{50}$$

$$10DE = 1000$$

$$DE = 100 \text{ m}$$

$\therefore$ Height of tower = 100 m (Option 1)

- ii If ~~the~~ Mural's house casts shadow of 15m, $CB = 15 \text{ m}$
~~so~~, $\triangle ACB \sim \triangle DEF$ (above)

$$\frac{AC}{DE} = \frac{CB}{EF} = \frac{AB}{DF}$$

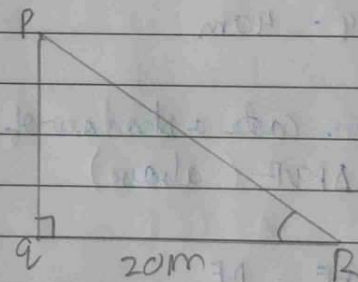
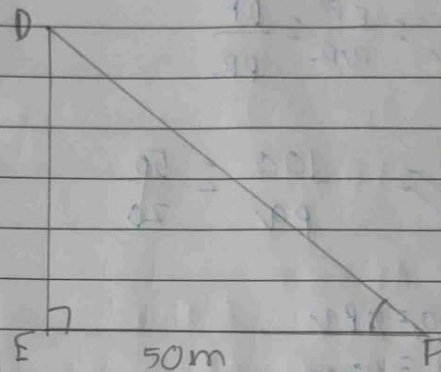
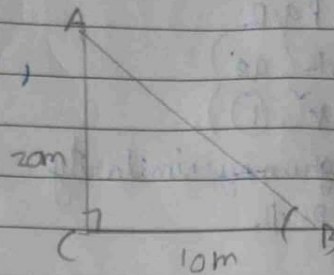
$$\frac{20}{100} = \frac{15}{EF}$$

$$20 \cdot EF = 1500$$

$$\therefore EF = 75 \text{ m}$$

Option 3 - 75 m

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iii

In $\triangle DEF$, PQR

$$\angle DEF = \angle PQR (90^\circ)$$

$$\angle DFE = \angle PRQ (1)$$

By AA ~~congruency~~ similarity

$$\triangle DEF \sim \triangle PQR$$

$$\frac{DE}{PQ} = \frac{EF}{QR} = \frac{DF}{PR}$$

$$\frac{100}{50} = \frac{100}{PQ} = \frac{50}{20}$$

$$200 = 5PQ$$

$$PQ = 40m$$

Option 4 - 40m

iv

If tower casts a shadow of 40m, $EF = 40m$
 $\triangle DEF \sim \triangle PQR$ (above)

$$\frac{DE}{PQ} = \frac{EF}{QR} = \frac{DF}{PR}$$

$$\frac{100}{40} = \frac{40}{QR}$$

$$10QR = 160$$

$$QR = 16m$$

$\therefore$ Option 2 - 16m

v

If tower casts a shadow of 40m, $EF = 40m$
 $\triangle ACB \sim \triangle DEF$ (above)

$$\frac{AC}{DE} = \frac{CB}{FE} = \frac{AB}{DE}$$

$$\frac{AC}{DE} = \frac{CB}{EF} = \frac{AB}{PF}$$

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$$\frac{20}{100} = \frac{CB}{40}$$

$$80 = 10CB$$

$$CB = 8 \text{ m}$$

∴ option 4 - 8 m

17 Let speed of boat in still water be x km/h

Speed of water flowing in river = y km/h

When boat going upstream, speed = 6 km/h

$$x - y = 6 \text{ km/h} \rightarrow (1)$$

When boat going downstream, speed = 14 km/h

$$x + y = 14 \text{ km/h} \rightarrow (2)$$

Solving (1) (2)

$$2x = 20$$

$$x = 10 \text{ km/h} \rightarrow (3)$$

Substitute in (1),

$$10 - y = 6$$

$$y = 4 \text{ km/h} \rightarrow (4)$$

i from (3),
option 2: 10 km/h

ii from (4)
option 2: 4 km/h

iii option 1 - Pair of linear equations

iv option 2 - Downstream

v $\frac{10+4}{2} = 7 \text{ km/h}$, Option 1: 7 km/h

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a

AB = Cable tower

CD = Building = 7m

Draw $DE \perp AB$ In $\triangle DCB$, $\angle C = 90^\circ$

$$\tan 45 = \frac{7}{BC}$$

$$7 = 1 \cdot BC$$

BC

$$BC = 7m \rightarrow (1)$$

$$BC = ED = 7m$$

~~Let $AE = x$ m~~In $\triangle AED$, $\angle E = 90^\circ$

$$\tan 60 = \frac{AE}{7}$$

$$AE = \sqrt{3} \cdot 7$$

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$$AE = 7\sqrt{3}$$

$$\text{Now } AB = AE + ED$$

$$= 7\sqrt{3} + 7$$

$$= 7(\sqrt{3} + 1) m$$

$$\therefore \text{Height of tower} = 7(\sqrt{3} + 1) m$$

b

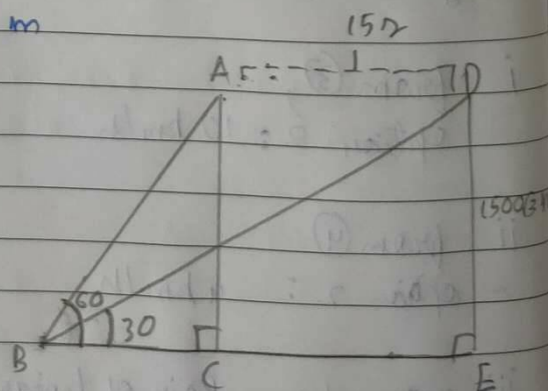
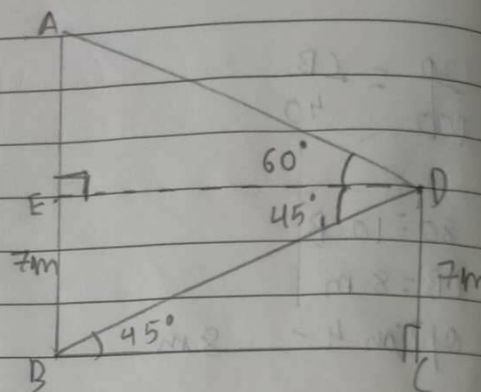
$$AC = DE = 1500\sqrt{3} m$$

In $\triangle ABC$, $\angle C = 90^\circ$

$$\tan 60 = \frac{AC}{BC}$$

$$\sqrt{3} = \frac{1500\sqrt{3}}{BC}$$

$$BC = 1500 m$$



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In $\triangle DBE$, $\angle E = 90^\circ$
 $\tan 30 = \frac{DE}{BE}$

$$\frac{1}{\sqrt{3}} = \frac{1500\sqrt{3}}{BC+CE}$$

$$BC+CE = 1500 \times 3$$

$$1500 + CE = 4500$$

$$CE = 3000 \text{ m}$$

$$\text{New speed} = \frac{\text{Distance}}{\text{time}} = \frac{CE}{15} = \frac{3000}{15} = \frac{2000 \text{ m/s}}{1} = 200 \text{ m/s}$$

$$\therefore \text{Speed of jet} = 200 \text{ m/s}$$

15

i $P(E) = \frac{2}{52} \boxed{\frac{1}{26}}$ (since there are 2 red kings)

ii $P(E) = \frac{12}{52} \times \frac{6}{26} \boxed{\frac{3}{13}}$ (since there are 12 face cards)

iii $P(E) = \frac{6}{52} \boxed{\frac{3}{26}}$ (since there are 6 red face cards)

iv $P(E) = \frac{1}{52} \boxed{\frac{1}{52}}$ (since there is only 1 jack of hearts)

v $P(E) = \frac{13}{52} \boxed{\frac{1}{4}}$ (since there are 13 spades)

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14 $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$

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13 given : $XY \parallel X'Y'$

AB is tangent with point of contact C

To prove : $\angle AOB = 90^\circ$

Proof: In $\triangle PAO$ and $\triangle CAO$

$\angle P = \angle C = 90^\circ$ (radius $\perp$ to tangent at point of contact)

$OP = OC$ (radii)

$PA = AC$ (tangents from same point)

By SAS congruency

$\triangle PAO \cong \triangle CAO$

$\angle POA = \angle AOC$ (C.P.C.T) $\rightarrow$ (1)

In $\triangle BCO$ and $\triangle BQO$

$\angle Q = \angle C = 90^\circ$ (radius $\perp$ to tangent at point of contact)

$OQ = OC$ (radii)

$QB = BC$ (tangents from external point)

By SAS congruency

$\triangle BCO \cong \triangle BQO$

$\angle COB = \angle QOB$ (C.P.C.T)

as PQ passes through centre O, it is a diameter, so it is also a straight line,

$$\angle POA + \angle AOC + \angle COB + \angle QOB = 180$$

$$2\angle AOC + 2\angle COB = 180$$

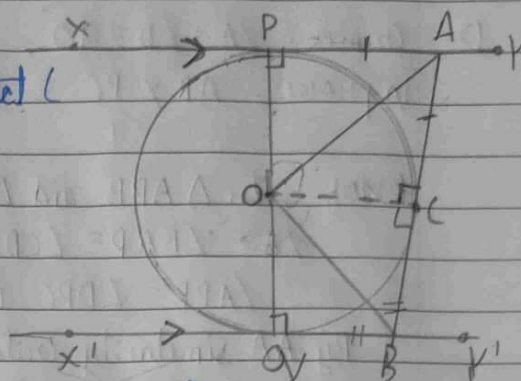
$$2(\angle AOC + \angle COB) = 180$$

$$\angle AOC + \angle COB = 90$$

$\therefore \angle AOC + \angle COB$ also equal to $\angle AOB$

Hence, $\angle AOB = 90^\circ$

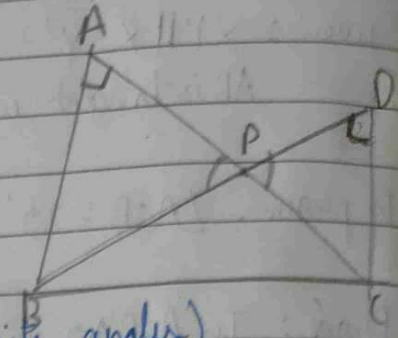
Hence proved.



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12 Given : $\angle A = \angle D = 90^\circ$
To prove : $AP \times PC = BP \times PD$



Proof : In $\triangle APB$ and $\triangle DPC$
 $\angle BAP = \angle CDP (90^\circ)$
 $\angle APB = \angle DPC$ (Vertically opposite angles)

By AA similarity criterion
 $\triangle APB \sim \triangle DPC$

$$\frac{AP}{DP} = \frac{PB}{DC} = \frac{AB}{BC} \quad (\text{corresponding parts of similar triangle})$$

$$\frac{AP}{DP} = \frac{PB}{DC}$$

$\therefore AP \times PC = PB \times PD$
Hence proved

11 $\frac{2}{x} + \frac{3}{y} = 2$, $\frac{1}{x} - \frac{1}{2y} = \frac{1}{3}$

~~$$\frac{2y + 3x}{xy} = 2$$~~

~~$$\frac{2y - x}{2xy} = \frac{1}{3}$$~~

~~$$2y + 3x = 2xy \rightarrow (1)$$~~

~~$$2y - x = \frac{2xy}{3} \rightarrow (2)$$~~

let us assume $\frac{1}{x} = p$ and $\frac{1}{y} = q$

so, $2p + 3q = 2$ ~~$\rightarrow (1)$~~ , $p = \frac{2 - 3q}{2} \rightarrow (1)$

$$p - \frac{q}{2} = \frac{1}{3} \rightarrow (2)$$

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substitute (1) in (2)

$$\frac{2-3q}{2} - \frac{q}{2} = \frac{1}{3}$$

$$\frac{2-4q}{2} = \frac{1}{3}$$

$$2-4q = \frac{2}{3}$$

$$-4q = \frac{2}{3} - 2$$

$$-4q = -\frac{4}{3}$$

$$-4q =$$

$$q = \frac{-\frac{4}{3} \times -1}{-4} \quad q = \frac{1}{3}$$

$$2p + 3q = 2$$

$$2p + 1 = 2$$

$$2p = 1$$

$$p = \frac{1}{2}$$

$$\therefore x = 2 \quad \text{and} \quad y = 3$$

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Substitute ⑩ in (2), $\frac{2-3q}{2} - \frac{q}{2} = \frac{1}{3}$

$$\frac{2-4q}{2} = \frac{1}{3}, \quad 2-4q = \frac{2}{3}$$

1 $1 : \frac{1}{2}$

2 ~~2 cm~~ $2 : 9 \text{ sq. units}$

3 $2 : 2 \text{ cm}$

4 $1 : 60^\circ$

5 $4 : \text{No solution}$

6 $1 : 2$

7 $2 : 1$

8 $3 : 7 - 4\sqrt{3}$

9 $4 : 10 \text{ cm}$

10 $2 : \frac{\sqrt{1+\cot^2 \theta}}{\cot \theta}$